# Alexander Montello

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## ENGINEERING PORTFOLIO:

alexmontello.github.io/portfolio

## PROFESSIONAL SUMMARY

Mechanical Engineering student at Purdue University with hands-on experience in design, analysis, and fabrication of electromechanical systems. Focused on developing manufacturable, high-performance mechanisms through rapid prototyping, testing, and iterative engineering considering tradeoffs and real-world constraints.

## EDUCATION

**Purdue University: Mechanical Engineering; 3.94 GPA,** West Lafayette IN — *AUGUST 2025 - MAY 2029*

- Led mechanical design for an instrument tuning tool for visually impaired musicians in a team of 8 students
- Used a PDM system to simulate an industry PLM setting in MFET 163 using ECOs and approval workflows in NX

## ENGINEERING EXPERIENCE

**Formula SAE, *Chassis Subteam Member*,** West Lafayette IN — *AUGUST 2025 - PRESENT*

- Produced manufacturing drawings with GD&T and used manual and CNC machines to manufacture 30+ parts
- Validated buckling and stiffness limits of pedal assembly using FEA with advanced meshing
- Designed and implemented a backup throttle cable mechanism and physical brake pedal structural validation test
- Assembled and tested a pneumatic shifting system, added an accumulator to reduce shift times by 76%

**FIRST Tech Challenge Robotics, *Mechanical Lead*,** City/State/International — *AUGUST 2022 - JUNE 2025*

- Designed and led manufacturing of hundreds of robot components, focusing on parallel motions and low inertia
- Rapidly iterated on designs using CAD version control, data analysis with Excel, and detailed documentation
- Won 8 awards across design, control, and our engineering process, including 2 at the international level

**See3D, *CAD and Printing Intern*,** Cincinnati OH — *JUNE 2025 - AUGUST 2025*

- Designed 30+ diverse 3D models optimized for 3D printing and accessibility using parametric and polygonal CAD
- Managed operation and maintenance of 10 FDM printers, maximizing uptime and troubleshooting issues
- Developed the first Braille CAD tool for Onshape for more efficient modeling and maintained design SOPs

## DESIGN PROJECTS

**Articulated Robotic Arm**

- Created a compact multi-DOF arm with worm-driven joints to prevent backdriving and centralize motor mass
- Implemented inverse kinematics for coordinated turret and arm positioning in a 700 mm radius work envelope
- Integrated a compliant differential wrist claw enabling robust and forgiving grasping across a large workspace

**Multilayer Rubik's Cube Solving Robot**

- Developed a novel machine with 6 modular grippers allowing decoupled linear and rotational motion for speed
- Created a clustering-based color detection pipeline and implemented collision-aware move scheduling
- Achieved 1.1 second move cycles and eliminated risk of gripper crashes using compliant materials and software

**Cable Driven Parallel Robot**

- Designed and built a six-cable parallel manipulator driven by six servo actuated pulley arms to maximize travel
- Programmed matrix-based inverse kinematics and polynomial regression mapping poses to actuator angles
- Achieved 0.9 m/s and 7.5 rad/s linear and angular speeds to move an effector around a 72600 $cm^3$ workspace

## TECHNICAL SKILLS

Onshape, Ansys Mechanical, Autodesk Inventor, Siemens NX/Teamcenter, Fusion 360 CAM, CNC milling, manual mills, manual lathes, band saws, power tools, MATLAB, KiCad, microcontrollers, electronics, Excel, Python, Java, C